Paper III
Gradient-Activated Phase Leakage and Vortex-Core Interiors in Coherence Gradient Gravity

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Abstract

This paper extends the nonlinear compact-object sector of Coherence Gradient Gravity (CGG) by introducing gradient-activated approximate U(1) phase leakage and vortex-core interiors. The General Relativity (GR) event horizon is preserved as the causal boundary. Exterior compact-object behavior remains Kerr-consistent. Interior strong-field dynamics transition into metastable vortex filaments under rotational phase pressure.

Comparative Framework: GR vs Paper II vs Paper III

Feature	General Relativity (GR)	Paper II (Smooth Branch CGG)	Paper III (Vortex-Core Extension)
Exterior Metric	Schwarzschild / Kerr exact solutions	Exterior identical to GR (smooth $\delta \geq 0$ basin)	Exterior identical to GR; vortices confined to interior
Event Horizon	Geometric null surface	Mode-transition surface; GR preserved outside	GR horizon retained; interior phase transition allowed
Interior Structure	Classical singularity	Smooth nonlinear coherence envelope	Metastable vortex filaments ($A \to 0$ cores)
Love Numbers (Static)	$k_2 = 0$ for black holes	Exponential suppression via curvature control	Suppression preserved; vortex behind horizon
Ringdown	Kerr quasi-normal modes	Smooth cavity-like nonlinear response	Kerr exterior + weak vortex Kelvin-mode coupling
Electromagnetic Behavior	Null geodesic trapping	Mode-sector evanescence near transition	Interior phase-gradient-induced attenuation
Topology / Phase	No internal phase	Complex field;	Approximate U(1)

	degree	smooth branch dominant	with gradient-activated leakage